Project Executive Summary
Predicting the SNAP Gap

GitHub: [Local Poverty Rate Estimation Project]

1.Project Overview

Build a predictive model to identify vulnerable households eligible for SNAP but not receiving it (the "SNAP Gap"). We use observable socioeconomic and demographic proxy features (e.g., household size, education, language barriers, internet access) to classify these households, enabling pure prediction for outreach tools

2. Motivation and Problem Statement

Official poverty statistics are often unreliable at small geographic levels due to limited sample sizes. This project addresses the challenge by building predictive models that estimate poverty rates using demographic and socioeconomic features.

3. Stakeholders

1) Federal Government Agencies:
    - Housing assistance
    - SNAP
    - welfare programs
2) Non-Profit Organizations and NGOs:
    - Food banks
    - Housing assistance programs
3) State and Local Governments
4) Policymakers and Congress :
    - Poverty reduction policies
    - Education and employment programs
5) Businesses and Private Sector
6) Emergency and Public Health Agencies
7) Urban Planners and Community Developers
8) Data Scientists & Analysts
9) Researchers benefit from accurate poverty estimates to design interventions, allocate funding, and evaluate programs.

4. Data Sources

The project uses ACS Public Use Microdata Sample (PUMS) data, including variables such as income, employment status, education, and household composition.

⇒ Datasets:
https://www.census.gov/programs-surveys/acs/microdata.html

5. Data Cleaning

The data cleaning process was a critical step to ensure the reliability and quality of the analysis. The project utilizes ACS Public Use Microdata Sample (PUMS) data, which contains detailed demographic and socioeconomic information but requires extensive preprocessing due to missing values, inconsistencies, and noise.

⇒ Data Selection and Filtering

Relevant variables were selected based on their importance for modeling poverty and socioeconomic status. Key features include income (PINCP), poverty ratio (POVPIP), employment status (ESR), education level (SCHL), age (AGEP), and household size (NP). The dataset was filtered to include individuals aged 18 and above to focus on the working-age population.

⇒ Handling Missing and Invalid Values

Records with missing or invalid values in critical variables were carefully handled. Observations with non-positive or unrealistic income values were removed to maintain data integrity. Additional filtering was applied to ensure consistency across key socioeconomic indicators.

⇒ Target Variable Construction

A binary poverty indicator was created using the poverty-to-income ratio (POVPIP). Individuals with POVPIP values below the defined threshold were classified as living in poverty, forming the primary target variable for modeling.

⇒ Feature Engineering

New variables were derived to better capture household and socioeconomic dynamics. These include aggregated household characteristics, employment indicators, and transformations of existing variables to improve model interpretability and predictive performance.

⇒ Encoding Categorical Variables

Categorical features, such as employment status and education level, were converted into numerical representations using one-hot encoding. This transformation ensures compatibility with machine learning algorithms.

⇒ Outlier Detection and Treatment

Extreme values, particularly in income, were identified and treated using percentile-based thresholds (e.g., 99th percentile). This step reduces the influence of outliers and improves model stability.

⇒ Data Preparation for Modeling

The cleaned dataset was standardized and structured for modeling. Features were scaled where necessary, and the dataset was split into training, validation, and test sets to support robust model evaluation.

6. Modeling Approach

To predict poverty status and identify households at risk of falling into the SNAP gap, this project employs a supervised machine learning framework using cleaned ACS data. Multiple classification models were evaluated to capture both linear and nonlinear relationships between socioeconomic variables and poverty outcomes.

⇒ Model Selection and Evaluation

The following models were implemented and compared:

- Logistic Regression (baseline model)
- Random Forest
- LightGBM
- XGBoost

Model performance was evaluated using standard classification metrics, including **AUC, precision, recall, and F1-score**, to ensure a balanced assessment between correctly identifying vulnerable populations and minimizing false predictions.

Model	AUC	Precision	Recall	F1
Logistic Regression	0.77	0.82	0.73	0.77
Random Forest	0.81	0.81	0.80	0.81
LightGBM	0.82	0.82	0.83	0.83
XGBoost	0.82	0.85	0.74	0.79

⇒ Model Performance and Selection

Among all evaluated models, LightGBM emerged as the best-performing model, achieving the highest overall performance across key metrics:

- AUC ≈ 0.82
- Precision ≈ 0.82
- Recall ≈ 0.83
- F1-score ≈ 0.83

LightGBM demonstrated a strong balance between precision and recall, making it particularly suitable for identifying vulnerable populations without excessively increasing false positives.

⇒ Final Evaluation

The final model was evaluated on the test dataset using a confusion matrix, which confirmed its effectiveness in distinguishing between households experiencing the SNAP gap and those that are not. The model achieved strong classification performance, correctly identifying a large proportion of true positive cases while maintaining reasonable accuracy for negative cases

7. Results

The modeling results demonstrate that machine learning approaches can effectively identify households at risk of experiencing the SNAP gap using socioeconomic and demographic features. Among the evaluated models, **LightGBM** achieved the best overall performance, outperforming Logistic Regression, Random Forest, and XGBoost across key evaluation metrics.

LightGBM produced the highest and most balanced scores, with strong performance in terms of AUC (≈ 0.82), precision (≈ 0.82), recall (≈ 0.83), and F1-score (≈ 0.83). This indicates that the model is capable of accurately identifying households that are likely to be eligible for SNAP but not receiving benefits, while maintaining a good balance between false positives and false negatives.

The confusion matrix further confirms the model's effectiveness, showing a high number of correctly classified cases for both gap and non-gap households. This reflects the model's ability to generalize well to unseen data and capture complex, nonlinear relationships within the dataset.

In addition to model performance, exploratory analysis highlights a critical finding: a significant portion of low-income households fall into the SNAP gap. These households meet eligibility conditions but are not enrolled in the program, revealing a disconnect between policy design and real-world participation.

Overall, the results demonstrate that:

- ⇒ LightGBM is the most effective model for predicting SNAP gap status
- ⇒ Socioeconomic variables strongly influence participation outcomes
- ⇒ A measurable and meaningful gap exists between eligibility and actual benefit receipt

8. Impact and Applications

The findings of this project carry significant implications for policymakers, government agencies, and nonprofit organizations dedicated to poverty reduction and social welfare programs. First, the model allows for precise identification of underserved populations, especially households that are likely eligible for SNAP but are not receiving benefits. This enables agencies to go beyond broad eligibility criteria and target individuals at the highest risk of being excluded from assistance. Second, identifying the SNAP gap offers actionable insights into systemic barriers to participation, such as limited access, lack of awareness, and administrative challenges. Addressing these obstacles can boost program participation and ensure help reaches those who need it most. Third, the framework promotes data-driven decision-making, helping stakeholders allocate resources more effectively and design interventions based on predictive insights rather than relying solely on aggregate statistics. Furthermore, this approach can be adapted to other public assistance programs, facilitating improved outreach strategies, enhanced program evaluation, and a fairer distribution of resources. Ultimately, this project shifts the focus from merely measuring poverty to understanding and closing gaps in assistance delivery, ensuring that eligible populations are reached and supported more effectively.

9. Limitations

Despite strong model performance, several limitations should be considered:

- **Survey Data Limitations:**
  ACS data is subject to sampling variability, measurement error, and missing values, which may affect model accuracy.
- **Small-Area Estimation Challenges:**
  Estimates at fine geographic levels may be unstable and require further validation or smoothing techniques.
- **Model Generalization:**
  The model is trained on a specific dataset and may not generalize equally well across different regions or time periods.
- **Lack of Direct SNAP Variable:**
  The dataset does not include a direct indicator of SNAP participation. As a result, SNAP receipt cannot be measured explicitly and must be inferred using proxy variables such as income level and the TARGET_GAP indicator.
- **Class Imbalance:**
  The distribution of poverty vs. non-poverty cases may introduce bias in model predictions, particularly affecting recall and precision.

10. Next Steps

Future Work Includes:

- Incorporating Additional Datasets.
- Improving SNAP Estimation Methods.
- Applying Advanced Modeling Techniques.
- Enhancing Geographic Analysis.
- Improving Model Interpretability.
- Validating The Model Across Regions.